Problem 3.1

Consider the accumulation function $a(t) = t^2 + t + 1$.

(a) Find the effective interest rate i.

(b) Find i_n.

(c) Show that i_n is decreasing.

Solution

(a) The effective interest rate is

$$i = \frac{a(1) - a(0)}{a(0)}.$$

Since

$$a(1) = 1^2 + 1 + 1 = 3, \qquad a(0) = 1,$$

we get

$$i = \frac{3-1}{1} = 2.$$

Therefore,

$$i = 200\%.$$

(b) By definition,

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)}.$$

Also,

$$A(n) = A(0)a(n), \qquad A(n-1) = A(0)a(n-1).$$

Hence,

$$i_n = \frac{A(0)a(n) - A(0)a(n-1)}{A(0)a(n-1)} = \frac{a(n) - a(n-1)}{a(n-1)}.$$

Now,

$$a(n) = n^2 + n + 1,$$

and

$$a(n-1) = (n-1)^2 + (n-1) + 1 = n^2 - n + 1.$$

Therefore,

$$i_n = \frac{(n^2 + n + 1) - (n^2 - n + 1)}{n^2 - n + 1} = \frac{2n}{n^2 - n + 1}.$$

(c) Let

$$f(x) = \frac{2x}{x^2 - x + 1}.$$

We differentiate to study whether i_n is decreasing:

$$f'(x) = \frac{2(x^2 - x + 1) - 2x(2x - 1)}{(x^2 - x + 1)^2}.$$

Simplifying the numerator,

$$f'(x) = \frac{2x^2 - 2x + 2 - 4x^2 + 2x}{(x^2 - x + 1)^2} = \frac{-2x^2 + 2}{(x^2 - x + 1)^2}.$$

Thus,

$$f'(x) = \frac{2(1 - x^2)}{(x^2 - x + 1)^2} < 0 \qquad \text{for } x > 1.$$

Hence $f(x)$ is decreasing for $x > 1$, and therefore

$$i_n = \frac{2n}{n^2 - n + 1}$$

is decreasing.